COMP3311 Week 1 Thursday Lecture

- Instructor's Intro
- CSE Environment
- ER Model (recap)
- Exercise: ER Models
- Class Hierarchies
- Class Hierarchies/Lecturer Class Hierarchy
- Developing ER Models
- Exercise: Medical Information
- Relational Model
- Relational Model
- Relational Data Model
- ER-to-Relational Mapping
- SQL Schemas
- Exercise: ER-to-SQL (I)
- Exercise: ER-to-SQL (II)
- Exercise: ER-to-SQL (III)

In today's lecture ...

Topics:

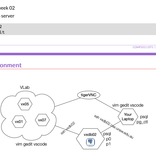
- Entity-Relationship (ER) Model (core)
- Relational Model
- Mapping ER → SQL
- SQL Data Definition Language (DDL)

Reminders:

- tutes start in week 02
- set up your PG server

```
$ ssh vxdb02
3311: pgsql
```


CSE Environment



ER Model (recap)

ER diagrams give a visual representation of data model

Primary components

- Entities = objects of interest in the scenario
- Relationships = associations between entities
- Attributes = properties of entities

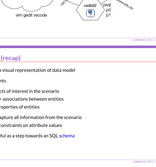
ER models don't capture all information from the scenario

- e.g. types and constraints on attribute values

ER models are useful as a step towards an SQL schema

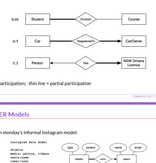
ER Model (recap) (cont)

ER design elements:



ER Model (recap) (cont)

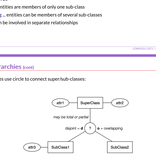
Relationships:



Thick line = total participation; thin line = partial participation

Exercise: ER Models

ER model based on monday's informal instagram model:



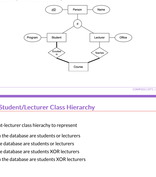
Class Hierarchies

ER also implements super-class / sub-class hierarchies

- both super- and sub-classes consist of entities
- super-class has common properties of all entities in hierarchy
- sub-classes can add extra properties to specialise
- entities in super-class may have corresponding entities in sub-class
- sub-classes can be
  - disjoint ... entities can be members of only one sub-class
  - overlapping ... entities can be members of several sub-classes
- sub-classes can be involved in separate relationships

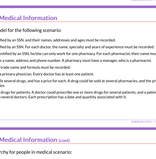
Class Hierarchies (cont)

ER class hierarchies use circle to connect super/sub-classes:



Class Hierarchies (cont)

Consider a concrete example of a class hierarchy:



Exercise: Student/Lecturer Class Hierarchy

Modify the student-lecturer class hierarchy to represent

- some people in the database are students or lecturers
- all people in the database are students or lecturers
- all people in the database are students XOR lecturers
- some people in the database are students XOR lecturers

Developing ER Models

A complete ER model should ...

- include all entities and their attributes
- identify key attributes
- capture accurate semantics for all relationships
- include class hierarchies where appropriate

There are frequently different kinds of people (→ sub-classes)

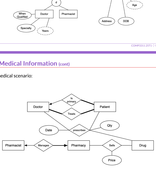
Exercise: Medical Information

Develop an ER model for the following scenario:

- Patients are identified by an SSN, and their names, addresses and ages must be recorded.
- Doctors are identified by an SSN. For each doctor, the name, specialty and years of experience must be recorded.
- A pharmacist is identified by an SSN, but he/she can only work for one pharmacy. For each pharmacist, their name must be recorded.
- Each pharmacy has a name, address and phone number. A pharmacy must have a manager, who is a pharmacist.
- For each drug, the trade name and formula must be recorded.
- Every patient has a primary physician. Every doctor has at least one patient.
- Each pharmacy sells several drugs, and has a price for each. A drug could be sold at several pharmacies, and the price could vary between pharmacies.
- Doctors prescribe drugs for patients. A doctor could prescribe one or more drugs for several patients, and a patient could obtain prescriptions from several doctors. Each prescription has a date and quantity associated with it.

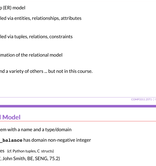
Exercise: Medical Information (cont)

Simple class hierarchy for people in medical scenario:



Exercise: Medical Information (cont)

Class hierarchy for people in medical scenario:



Exercise: Medical Information (cont)

Relationships in medical scenario:

Exercise: Medical Information (cont)

Prescriptions in medical scenario:

Types of Data Models

Entity-relationship (ER) model

- world is modelled via entities, relationships, attributes

Relational model

- world is modelled via tuples, relations, constraints

SQL schemas

- a good approximation of the relational model

Also OOL, UML, and a variety of others ... but not in this course.

Relational Model

Attribute = data item with a name and a type/domain

- e.g. **account_balance** has domain non-negative integer

Tuple = list of values (cf. Python tuples, C structs)

- e.g. (1234567, John Smith, BE, SENG, 75.2)

Relation = set of tuples

- e.g. {(1,2,3), (3,2,1), (1,3,5), (2,4,6) }

Constraint = logical statements on valid data

- e.g. zID is unique and 0 ≤ WAM ≤ 100

Relational Model (cont)

Correspondence between ER and Relational models:

- Relational attributes correspond to ER attributes
  - although ER attributes generally don't have explicit domains
- Relational tuples correspond to ER entities
- Relations correspond to sets of ER entities
- Relations also correspond ER relationships

Relational Data Model

A relational schema consists of

- a collection of relation definitions + constraints

Different kinds of constraints

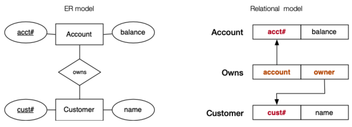
- unique = value of attribute is unique in relation
- key = chosen unique attribute to distinguish tuples
- domain = type of attribute, restrictions within type
- referential integrity = foreign key
  - tuple in relation R has attribute F
  - whose value corresponds to key attribute K in relation S

❖ Relational Data Model (cont)

Mapping an ER model to a relational schema

- attributes → attributes, plus domains plus constraints
- entities → tuples, entity sets → relations
- relationships → relations, plus constraints

Example:



❖ ER to Relational Mapping

Above example gives informal description of relational schema

Need a more formal way of describing relational schemas

SQL data definition language (DDL) provides this

SQL as an implementation of relational data model

- relations → tables, tuples → tuples, attributes → columns/fields

❖ ER to Relational Mapping (cont)

Useful strategy for database design:

- perform initial data modelling using ER
  (conceptual-level modelling, gives a "map" of the DB)
- transform conceptual design into SQL relational model
  (implementation-level modelling)

A formal mapping exists for ER model → SQL/Relational model.

This maps "structures"; but additional info is needed, e.g.

- concrete domains for attributes and other constraints

But can't map some things (e.g. n:m total participation)

❖ SQL Schemas

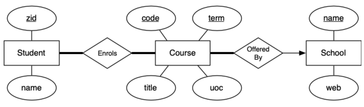
Primary SQL DDL construct is table creation:

```
create table TableName (
   attr1Name  type [constraints],
   attr2Name  type [constraints],
   attr3Name  type [constraints],
   ...
   primary key (attrxName ),
   foreign key (attryName)
               references OtherTable (attrzName )
);
```

SQL schema = collection of table definitions, including constraints

❖ Exercise: ER to SQL (i)

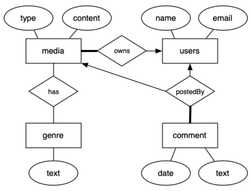
Convert the following ER diagram to an SQL schema:



❖ Exercise: ER to SQL (ii)

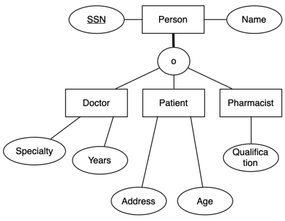
Convert the following ER diagram to an SQL schema:

```
Instagram data model

objects
media: photos, videos
users:name
label:text
comment:date,text,..


attributes
media: genre(s)


relationships:
media-ownedby-user
media-has-label
media-has-comment
comment-postedBy-user
```



❖ Exercise: ER to SQL (iii)

Convert the following ER diagram to an SQL schema:



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